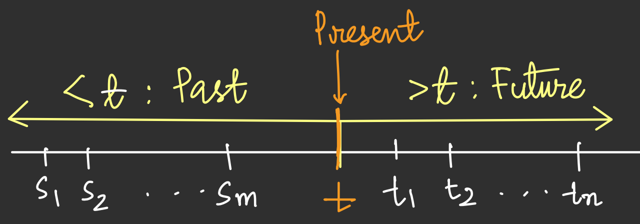
EE2080 : Lecture 8 : 18 Aug 2026



RECAP : 1) The Markov Property:

For an RP $\{X_t : t \in T\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\{X_{s_1}, X_{s_2}, \dots, X_{s_m}\} \perp\!\!\!\perp \{X_{t_1}, X_{t_2}, \dots X_{t_n}\} \mid X_t \text{ , where}$$

$$s_1 < s_2 < \dots < s_m < t < t_1 < t_2 < \dots < t_n$$

Given the present, the future is independent of the past!

2) Discrete time Markov Chain

- $\{X_n, n \in \mathbb{N}\}$, with X_n taking values in a countable set S (State space) s.t $\{X_n, n \in \mathbb{N}\}$ has the Markov property
- Transition Probability matrix $P(n) = |S| \times |S|$ matrix with the $(i,j)^{th}$ entry
  $P_{i,j}(n) = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}$, $n \in \mathbb{N}$, $i, j \in S$. (Row-Stochastic Matrix)
- Time Homogeneous DTMC $P(n) = P \;\; \forall\, n \in \mathbb{N}$
- n-step transition probabilities $\mathbb{P}\{X_{n+1} = j \mid X_1 = i\}$.

PLAN: 1) Chapman-Kolmogorov Eqn - Continued
2) Stopping time & Strong Markov Property
3) Hitting & Recurrence Times. & 4) Applications

Chapman - Kolmogorov Equation

Theorem (Chapman-Kolmogorov): For a time-homogeneous DTMC $\{X_n : n \in \mathbb{N}\}$ with state space S, for any $n \in \mathbb{N}$, let $P_{i,j}^{(n)} = \mathbb{P}\{X_{n+1} = j \mid X_1 = i\}$, $i, j \in S$. Let $P^{(n)} = [P_{i,j}^{(n)}]_{i,j \in S}$. Then, $\boxed{P^{(n)} = P^n \;\; \forall\, n \in \mathbb{N}}$, where $P = P^{(1)}$ is the one-step TPM of $\{X_n, n \in \mathbb{N}\}$

Proof: For $i, j \in S$ and $n, l \geq 0$

$$p_{i,j}^{(n+l)} = \mathbb{P}\{X_{n+l+1} = j \mid X_1 = i\}$$

$$= \sum_{k \in S} \mathbb{P}\{X_{n+1} = k \mid X_1 = i\} \cdot \mathbb{P}\{X_{n+l+1} = j \mid X_1 = i, X_{n+1} = k\}$$

$$= \sum_{k \in S} \mathbb{P}\{X_{n+1} = k \mid X_1 = i\} \, \mathbb{P}\{X_{n+l+1} = j \mid X_{n+1} = k\}$$

$$= \sum_{k \in S} p_{i,k}^{(n)} \cdot p_{k,j}^{(l)}$$

$P^{(n)}$ ith row $*$ $P^{(l)}$ jth coln.

$$= \left[P^{(n)} \cdot P^{(l)}\right]_{i,j}$$

$$P^{(n+l)} = P^{(n)} \cdot P^{(l)}.$$

Use this inductively to get the desired result.

Theorem (Chapman-Kolmogorov II)

Let $\{X_n : n \in \mathbb{N}\}$ be a TH-DTMC on the state space S with TPM P.
For each $n \in \mathbb{N}$, let $\pi(n)$ denote the unconditional PMF of X_n.

$$\boxed{\pi(n+1) = \pi(n) \cdot P.} \quad \forall n \in \mathbb{N} \qquad S = \{1, 2, \ldots, |S|\}.$$

$$\begin{bmatrix} \pi_1(n) & \pi_2(n) & \ldots & \pi_{|S|}(n) \end{bmatrix} \begin{bmatrix} \begin{pmatrix} P_{1,1} \\ P_{2,1} \\ \vdots \\ P_{|S|,1} \end{pmatrix} & & \end{bmatrix}$$

$$\pi_1(n) P_{1,1} + \pi_2(n) \cdot P_{2,1} + \ldots + \pi_{|S|}(n) \cdot P_{|S|,1}$$

$\mathbb{P}(X_n = 1)$ $\mathbb{P}(X_{n+1} = 1 \mid X_n = 1)$

$\mathbb{P}(X_n = 2) \cdot \mathbb{P}(X_{n+1} = 1 \mid X_n = 2)$

$$\sum P(X_n=i)\cdot P(X_{n+1}=1 \mid X_n=i) = P(X_{n+1}=1)$$

$\{X_n\}_{n\in\mathbb{N}}$.

$$\pi(2) = \pi(1)\cdot P.$$
$$\pi(3) = \pi(1)P^2.$$
$$\vdots$$
$$\pi(n) = \pi(1)\cdot P^{n-1}$$

<u>Finite Dimensional Distributions.</u>

FDDs of a TH-DTMC $\{X_n, n\in\mathbb{N}\}$ on the state space S.

$$P(X_{n_1}=i_1, X_{n_2}=i_2, \ldots, X_{n_m}=i_m) = \sum_{i\in S} P(X_1=i, X_{n_1}=i_1, \ldots X_{n_m}=i_m)$$

$1 < n_1 < n_2 < \ldots < n_m$.

$$= \sum_{i\in S} P(X_1=i)\cdot P(X_{n_1}=i_1 \mid X_1=i) P(X_{n_2}=i_2 \mid X_{n_1}=i_1) \ldots P(X_{n_m}=i_m \mid X_{n_{m-1}}=i_{m-1})$$

$$= \underbrace{\sum_{i\in S} \pi_i(1)\, P_{i,i_1}^{(n_1-1)}}\cdot P_{i_1,i_2}^{(n_2-n_1)} \cdots P_{i_{m-1},i_m}^{(n_m-n_{m-1})}$$

FDDs are completely specified using the TPM P and the initial PMF $\pi(1)$.

$$P(X_{n_1}=i_1, \ldots, X_{n_m}=i_m) = \underbrace{P(X_{n_1}=i_1)}\cdot \underbrace{P(X_{n_2}=i_2 \mid X_{n_1}=i_1)}_{P_{i_1,i_2}^{n_2-n_1}} \cdots \underbrace{P(X_{n_m}=i_m \mid X_{n_{m-1}}=i_{m-1})}_{P_{i_{m-1},i_m}^{n_m-n_{m-1}}}$$

$$P_{i_1,i_2}^{n_2-n_1}\; P_{i_2,i_3}^{n_3-n_2} \cdots P_{i_{m-1},i_m}^{n_m-n_{m-1}}$$

$$\underline{\pi(n_1)} = \underline{\left[\pi(1)\cdot P^{n_1-1}\right]_{i_1}}$$
$$\sum_{i\in S} \pi_i(1) P_{i,i_1}^{n_1-1}$$

$P(X_1 = i_1, \dots, X_n = i_n) \overset{?}{=} P(X_{m+1} = i_1, \dots X_{m+n} = i_n)$ for any $m \in \mathbb{N}$.

TH-DTMC with TPM P on state space S.

$$\underbrace{\pi_{i_1}(1) \, P_{i_1, i_2} \cdot P_{i_2, i_3} \cdots P_{i_{n-1}, i_n}}_{LHS} \qquad \underbrace{\pi_{i_1}(m+1)}_{\boxed{[\pi(1) \cdot P^m]_{i_1}}} \cdot P_{i_1, i_2} \cdot P_{i_2, i_3}, \dots, P_{i_{n-1}, i_n} \quad RHS.$$

TH-DTMC is not stationary!

$$P\left(\bigcap_{j=2}^{n} \{X_j = i_j\} \,\middle|\, \{X_1 = i_1\}\right) = P\left(\left\{\bigcap_{j=2}^{n} X_{m+j} = i_j\right\} \,\middle|\, \underbrace{\{X_{m+1} = i_1\}}\right)$$

$$LHS = P_{i_1, i_2} \cdot P_{i_2, i_3} \cdots P_{i_{n-1}, i_n} = P_{i_1, i_2} P_{i_2, i_3} \cdots P_{i_{n-1}, i_n}$$

Conditioned on the initial state, any finite dimensional joint PMF of a TH-DTMC $\{X_n, n \in \mathbb{N}\}$ is stationary; i.e $\forall \, m, n \in \mathbb{N}$ and $i_1, i_2, \dots i_n \in S$

$$P\left(\bigcap_{j=2}^{n} \{X_j = i_j\} \,\middle|\, \{X_1 = i_1\}\right) = P\left(\left\{\bigcap_{j=2}^{n} X_{m+j} = i_j\right\} \,\middle|\, \underbrace{\{X_{m+1} = i_1\}}\right)$$

In particular for a TH-DTMC $P\{X_{k+1} = j \mid X_1 = i\} = P\{X_{k+m} = j \mid X_m = i\}$

(k-step transition probabilities). $\qquad \forall \, i, j \in S, \, k, m \in \mathbb{N}$.

Strong Markov Property:

Example: Fix a prob. space $(\Omega, \mathcal{F}, P)$. Consider a TH-DTMC $\{X_n : n \in \mathbb{N}\}$ on the state space S. For $j \in S$, let T_j be the random time when the process first visits the state j, i.e $T_j(\omega) = k$ if $X_i(\omega) \neq j \;\; \forall \, 1 \le i \le k-1$ and $X_k(\omega) = j$. If for some $\omega \in \Omega$, $\nexists \, k$ s.t $T_j(\omega) = k$, then $T_j(\omega) = \infty$. Is T_j a RV? Is $P(X_{T+k} = j \mid X_1 = i_1, \dots, X_T = i) = P_{i,j}^{(k)}$?

$$\{\omega \in \Omega : T_j(\omega) = k\} = \left\{\omega \in \Omega : X_i(\omega) \neq j \ \forall\, 1 \le i \le k-1, \ X_k(\omega) = j\right\}$$

Is $\mathbb{P}\left(X_{T+k} = j \mid X_1 = i_1, \dots, X_T = i\right) = p_{i,j}^{(k)}$? Prove!

$$\mathbb{P}\left(X_{T+k} = j \mid X_1 = i_1, \dots, X_T = i\right) = \sum_{t=1}^{\infty} \mathbb{P}\left(X_{T+k} = j \mid X_1 = i_1, \dots, X_T = i, T = t\right) \mathbb{P}\left(T = t \mid X_1 = i_1, \dots, X\right)$$

$$= \sum_{t=1}^{\infty} \underbrace{\mathbb{P}\left(X_{t+k} = j \mid X_1 = i_1, \dots, X_t = i\right)}\,\mathbb{P}(T = t)$$

<u>Example</u>: TH-DTMC $\{X_n, n \in \mathbb{N}\}$ on the state space S with TPM $\mathbb{P}$.

$U^{(j)}$ – time of first return to state j. (time of second visit to j)

Let $T = U^{(j)} - 1$ (one step before the second visit to state j).

$$\mathbb{P}\left(X_{T+1} = k \mid X_1 = i_1, \dots, X_T = i\right) = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{P}\left(X_{T+1} = k \mid X_1 = i_1, \dots, X_T = i\right) = \mathbb{P}\{X_{T+1} = k \mid X_T = i\}$$

i.e $X_{T+1} \perp X_0, \dots, X_{T-1} \mid X_T$ <u>(Markov Property)</u>

Is $\mathbb{P}\{X_{T+1} = k \mid X_T = i\} = P_{i,k}$? No!

<u>Stopping Time</u>: A random time T is a stopping time for the random process $\{X_n, n \in \mathbb{N}\}$ on the state space S, if $\forall\, n \in \mathbb{N}$, $\exists$ a function $f_n : S^n \to \{0,1\}$ s.t $\mathbb{I}_{\{T \le n\}}(\omega) = f_n\left(X_1(\omega), \dots, X_n(\omega)\right)$, i.e to answer the question whether $T(\omega) \le n$ or not we only need to look at $X_1(\omega), \dots, X_n(\omega)$

For a TH-DTMC, the question of whether Markov property holds w.r.t a random time T is about

i) whether the RVs are Markovian or not.

ii) whether the transition probabilities are homogeneous & governed by the entries in the TPM P.

Is $\mathbb{P}\{X_{T+k}=j \mid X_1=i_1,\dots,i_T=i\} = \mathbb{P}\{X_{T+k}=j \mid X_T=i\} = (P^k)_{i,j}$?

Theorem: Fix a prob. space $(\Omega,\mathcal{F},P)$. Let $\{X_n, n\in\mathbb{N}\}$ be a TH-DTMC on state space S & let T be a stopping time s.t $\mathbb{P}\{T<\infty\}=1$. Then the DTMC has the strong Markov property w.r.t T, i.e.,

$$\mathbb{P}\{X_{T+k}=j \mid X_1=i_1, X_2=i_2,\dots,X_T=i\} = (P^k)_{i,j}$$

Proof: $P(X_1=i_1, X_T=i, X_{T+k}=j) = \sum_{t=1}^{\infty} P(X_1=i_1,\dots,X_t=i, X_{T+k}=j, T=t)$

$$= \sum_{t=1}^{\infty} \mathbb{P}(X_1=i_1,\dots,X_t=i)\, P(\underset{\uparrow \text{ fixed time, not random}}{X_{t+k}}=j \mid X_1=i_1,\dots,X_t=i)\cdot \mathbb{P}(T=t \mid X_1=i_1,\dots,X_{t+k}=j)$$

$$= \sum_{t=1}^{\infty} \underbrace{\mathbb{P}(X_1=i_1,\dots,X_t=i)\cdot \mathbb{P}(X_{t+k}=j \mid X_t=i)\cdot \mathbb{P}(T=t \mid X_1=i_1,\dots,X_t=i)}$$

$$= P^k_{i,j} \underbrace{\sum_{t=1}^{\infty} P(X_1=i_1,\dots X_t=i, T=t)}$$

$$= P^k_{i,j}\, \underbrace{P(X_1=i_1,\dots\dots,X_T=i)}_{>0}$$

$$P(X_{T+k}=j \mid X_1=i_1,\dots,X_T=i) = (P^k)_{i,j}$$